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Lactococcus lactis I7 isolated from traditional Italian cheese making: a biotechnological integrated platform

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Abstract

Background There is growing interest in newly isolated lactic acid bacteria from traditional sources. In this study, a *Lactococcus lactis* strain identified and isolated from natural whey starter cultures of cow milk for the production of artisanal cheeses was cultivated in optimized vegan grade medium to assess its growing ability and metabolic fingerprint. In fact, traditionally fermented dairy products are considered nutritionally complete and possibly functional to human health with a number of benefits not only to the gastro-intestinal tract but also in a systemic manner.

Results In fed batch experiments, we achieved 14 g/L dry cell weight and $1.9 \cdot 10^{10}$ viable colony forming unit increasing the values of experiment threefold and twofold respect to the batch processes, respectively. Additionally, the lactic acid (LA) production was quantified, and a maximal concentration of 78 g/L was achieved, which is approximately five-fold corresponding to the batch experiment results. This is in agreement with kinetic modeling of LA inhibition studies that highlighted a halved growth rate (μ) at 35 ± 5 g/L of LA whilst the growth blockage occurred at about 80 g/L. The samples obtaining after ultrafiltration processes and tested on three different pathogens, *Enterococcus faecalis*, *Salmonella enterica* subsp. *enterica* serovar *Typhimurium* and enteroinvasive *Escherichia coli*, gave a reduction on viability by 6–9 log on average.

Conclusion This paper focused its attention on optimization of fermentation conditions (in particular a vegan grade media using Design of Experiment approach) using fed-batch and microfiltration processes increasing the production of biomass and bioactive molecules, with respect to the batch processes. At the end of the fed-batch fermentation, a downstream process based on membranes was performed in order to obtain bioactive molecules that proved antimicrobial activity against intestinal/food spoilage pathogens.

Keywords Vegan grade medium optimization, Starter fermentation, Antimicrobial activity, Design of experiments, Lactic acid production

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Background

Lactococcus lactis and *Streptococcus thermophilus* strains are among the most used commercial starter cultures in the dairy industry for cheese ripening and manufacturing fermented milk. Recently, there has been growing interest in addressing their ability to produce bioactive molecules that can enrich the fermented food products, eventually becoming functional food. Some strains show health benefits and thus potentially they may be used in nutraceuticals and food supplements [1, 2].

Most lactic acid bacteria (LAB) belonging to the genera *Lactococcus* have been detected in various environments. For example, *L. lactis* strains were isolated mainly from fermented raw milk and kefir [3–5]. Their key function during milk fermentation is the rapid conversion of lactose into LA, which prevents the growth of pathogens in the fermented dairy product and also contributes to the final texture (moisture, softness) and taste [6]. Besides organic acids (lactic or other short chain fatty acids), a few LAB strains produce other metabolites, such as bacteriocins, that when released in the food processing may contribute to reduce spoilage thus extending the shelf life of food products [7]. In addition to the extensive use of bacteriocins for food preservation, some strains of *L. lactis* can also be useful to obtain starter cultures with protective properties and/or additional cultures to accelerate cheese maturation [1]. For a successful starter cultures, it is important to promptly increase biomass both to avoid potential contaminations and to reduce cost of process [8]. In fact, LA biosynthesis and lowering of pH is important in providing inhibitory effect for growth of other pathogenic organisms [9]. Besides LA, the interest is also focused on bacteriocins as food biopreservatives, that recalled great interest in recent years [10]. For this reason, beside the experiments related to medium and fermentation process design to achieve high biomass yield, to assess the optimal process to obtain the starter cultures at compatible costs and using selected ingredients, research activity is aimed at finding out whether this recently isolated *L. lactis* I7 produces bacteriocins at higher biomass density and LA concentrations. Malvido and collaborators performed studies including use of whey for batch and fed-batch experiments for the growth of *L. lactis* CECT 53 strain for bacteriocin production [11].

Fermentation process development using kinetic model may provide predefined path for fermentation progression which can be described in process design space. The knowledge of fermentation kinetics can help to have improved process control strategy in order to optimize productivity, which is also part of “Control strategy - Quality by Design” aspect of bioprocess development [12, 13].

Fed-batch process yields higher biomass production along with consistency in productivity compared to batch processes resulting in reduction in overall production cost. Thus, for starter cultures, it is vital to have stable fed-batch process developed [14]. Due to its simplicity and ease of control, less capital investment requirement, fed-batch fermentation frequently uses pulsed feeding as a general mode of operation [15, 16]. Enhancing feeding practices can boost cell biomass concentration during fermentation and increase productivity, which lowers production time and cost [17]. Numerous studies have concentrated on using fermentation to increase the production of probiotic biomass by utilizing in situ product removal bioreactors that stop LA buildup and, consequently, avoids growth inhibition [18]. Microfiltration batches are more demanding processes but delivers higher biomass density production. In order to increase the final microorganisms’ density and the production of bacteriocins a growth medium using a design of experiment (DoE) was optimized.

The vegan grade optimized medium was used to grow the microorganism following each of the fermentation strategies under investigation in order to better understand the best combination of nutrients to obtain a high yield of biomass and supernatant rich in bioactive molecules usable for the antimicrobial activity [19].

Furthermore, biomass production and the bacteriocin activity were improved through exploiting modern technologies of high cell density fermentations [20, 21]. Different fed batch strategies were tried. In particular, batch data on yield and glucose consumption were used to implement a fed-batch profile based on continuous feeding of substrate into bioreactor.

Several studies have focused their attention on optimization of fermentation conditions and on increasing the production of some LAB biomass via fermentation, with in situ product removal (ISPR) bioreactors that prevent accumulation of LA, and therefore growth inhibition [18, 22]. In fact, there are many papers reporting fed-batch fermentation processes designed so that the accumulation of inhibitory metabolites can be avoided or reduced thus resulting in higher biomass production [23]. However, the use of fed-batch fermentations even at controlled pH are often inefficient due to the high quantity of acids obtained. Therefore, to reduce the inhibitory effect of LA during the fermentation process, ISPR can be used [24]. These processes benefit from product removal during fermentation, hence the name in situ product removal, leading to an increase of biomass and related metabolites, improving yield and productivity and resulting in reduced process costs.

So, in this study, a full range experimental approach was accomplished to study the physiology of the newly isolated microorganism, namely *L. lactis* I7, optimize

fermentation medium, achieve high cell density through fed-batch and in situ product removal technique (high cell density culture, HCDC) and design a downstream based on ultrafiltration and diafiltration (UF/DF) to concentrate specific (molecular weight MW) metabolites (possibly bacteriocin) with antimicrobial activity. In fact, this concentrated fraction is highlighted in tests on human-enteric pathogens, frequently isolated in spoiled food. By studying different fermentation strategies, it was possible to identify the best solution in terms of biomass production and metabolites of interest to be used in the nutraceutical and food fields.

Methods

Bacterial strain

L. lactis subspecies *lactis* I7, used in the study, was previously isolated from natural whey starters cultures (it was the initial fermented milk) of cow milk for the production of artisanal cheeses [25]. *L. lactis* I7 was initially grown in M17 media in exponential growth phase followed by harvesting biomass in fresh sterile M17 media with 20% glycerol and preserving immediately at -80°C . The cryopreservation was carried in 1 ml stock with biomass optical density of 20 in 2 ml sterile and labelled Eppendorf tubes. After every 6 months, stocks were prepared again and older stocks were discarded.

Media components

The nitrogen sources were wheat peptone E1 (Organotechnie, France), neutralized soy peptone (Organotechnie, France), yeast extract (Organotechnie, France), rice peptone (Organotechnie, France) and guar peptone G01 (Organotechnie, France). Glucose, lactose, ascorbic acid and magnesium sulphate (MgSO_4) were provided by Sigma-Aldrich, Milano, Italy. M17 medium (containing 5 g/L of lactose, meat extract 5 g/L, meat peptone 2.5 g/L, soya peptone 5 g/L, tryptone 2.5 g/L, yeast extract 2.5 g/L, ascorbic acid 0.5 g/L, magnesium sulphate 0.25 g/L, and sodium glycerophosphate 19 g/L) was provided by Millipore (Switzerland). Salts used were di-sodium hydrogen phosphate (Na_2HPO_4) (SAFC, Switzerland), di-potassium hydrogen phosphate (K_2HPO_4)

(Sigma-Aldrich, Italy), di-sodium glycerophosphate ($\text{C}_3\text{H}_7\text{Na}_2\text{O}_6\text{P}$) (AppliChem, Germany).

Small scale experiments for screening of modified M17 medium sources through DoE

The experiments were performed in 50 ml sterile falcon tubes with 40 ml of medium, to leave just a low head air volume. Fermentation was carried out at 30°C with stirring of 150 rpm for 20 h in a shaker incubator (LAB Companion, WVR, Italy). The effects of M17 medium components and their concentrations on biomass production of *L. lactis* I7 were evaluated using medium compositions explained in Table 1. All experiments were performed in duplicate. To make a feasible number of the experiments and to gain a clear idea about which component is positively or negatively correlated to the biomass titer, 42 runs were considered sufficient to perform the screening of these five factors. Optical density was measured using DLab SP-UV1100 spectrophotometer at 600 nm against specific blanks (e.g. uninoculated medium). Small aliquot of samples was withdrawn and centrifuged at 6000 g to harvest biomass. The biomass was washed with 0.9% NaCl saline buffer 2 times to remove dissolved media components from pellet. The pellet was dried at 50°C for 48 h, followed by measurement of pellet weight for total dry cell weight. The supernatant was preserved at 4°C and successively further purified with 3KDa centricon tubes at 12,000 rpm. These samples were analysed in HPLC (UHPLC Dionex Ultimate 3,000; Thermofisher) on an Alltech IOA-2000 column (500 mm $\times$ 6.5 mm ID) at a flow rate of 0.6 ml/min. The mobile phase consisted of 0.1% H_2SO_4 in H_2O v/v. Peak areas were evaluated through the Thermofisher chromeleon software and quantified using external standard calibration curves. Standard concentrations were linear from 30 to 0.156 mg/ml for glucose, lactose, LA, acetic acid, and ethanol.

Screening of carbon source with one factor at a time (OFAT) in shake flask/bottles experiments

The effects of carbohydrates (namely glucose and lactose) at the same concentrations on biomass production were evaluated. All experiments were performed in duplicates, using the same conditions (volume 40 ml and temperature 30°C) described above.

Screening of salts with one factor at a time (OFAT) in shake flask/bottles experiments

The experiments were performed in 100 ml sterile glass bottles with 80 ml of medium, to have same head air volume. Fermentation was carried out at 30°C with stirring of 150 rpm for 20 h in a shaker incubator (LAB Companion, WVR, Italy). The effects of salt components at same concentrations on biomass production of *L. lactis*

Table 1 Medium component screening on small scale – various factor setting value in 2 levels with a mid-point with glucose constant at 20 g/L

Factors	code	Setting value		
		-1	0	+1
soy peptone	A	0	5	10
tryptone	B	0	5	10
yeast extract	C	0	5	10
ascorbic acid	D	0	0.25	0.5
MgSO_4	E	0	0.125	0.25

I7 were evaluated using medium compositions explained in Table 2. All experiments were performed in duplicates. The molecular concentration of the phosphate group was kept constant in all the experiment.

Lactic acid tolerance (substrate inhibition) in shake flask/ bottles experiments

The experiments were performed in 50 ml sterile falcon with 40 ml of medium, to have same head air volume in duplicates. Fermentation was carried out at 30 °C with stirring of 150 rpm for 5 h in a shaker incubator. The effects of the product (LA) at different concentrations (0 g/L, 10 g/L, 20 g/L, 30 g/L, 40 g/L, 50 g/L, 60 g/L, 70 g/L, 80 g/L, 90 g/L and 100 g/L) on biomass production of *L. lactis* I7 were evaluated. pH of all experimental runs were adjusted to 6.4±0.1 before and after sterilization of broths. Initial inoculum volume was calculated to have optical density of 0.5±0.10 at initial stage. All experiments were performed in duplicates.

The empirical data were processed to obtain plots and identify the maximum specific growth rate achieved in each run. The graph of maximum specific growth rate vs. initial LA concentration was also plotted. MATLAB's plot tool was used to identify various potential fittings for the available data. Custom equation plot tools was also used to check fitting of previously reported mechanistic and empirical models. The results were stored in jpeg format graph plot images. LA inhibition of *L. lactis* I7 was studied keeping concentrations constant. Samples were taken every 30 min for 2 h, and then at 3 h and 4 h, to measure optical density (O.D.) at 600 nm and thus quantify/calculate growth rate in the specific intervals. The initial pH of all media was adjusted to 6.4±0.1 and then the flasks were inoculated. Some results are also illustrated in the supplementary materials, Table S3.

Equation 1

$$\mu = \frac{\mu_{max}}{1 + e^{\lambda (x-x_0)}}$$

Equation 2

$$\mu = \mu_{max} \left(1 - \frac{x}{c}\right)^n$$

Where,

μ is specific growth rate

μ_{max} is maximum specific growth rate (0.9635)

λ is lactic acid product inhibition constant (0.08)

x is lactic acid concentration

x_0 indicates concentration of lactic acid with half maximum specific growth rate (35.1748 g/L)

c is maximum lactic acid concentration at which bacterial growth rate is almost zero.

Batch, batch with pulse and fed-batch processes

The bioreactor used for the experiments was a Biostat CT plus, Sartorius Stedim (Melsungen, Germany), 2 L working volume, 3.2 L total volume. *L. lactis* I7 was grown at temperature of 30 °C and 150 rpm, without gas. A constant pH of 6.5 was maintained with the addition of ammonium hydroxide and sulfuric acid [26]. Experiments in batch mode were carried out using the optimized medium on small scale in order to better understand the consumption of carbon source and the production of LA.

Batch with pulse and fed-batch experiments were performed with the same parameters of batch processes. In both types of experiments, feed was added when the concentration of glucose initially present in the reactor (20 g/L) was below 1±0.5 g/L. The glucose concentration in the feed was calculated by keeping total final glucose concentrations in final volume at 80 g/L. In the first case, the glucose concentration was restored two times by adding a single pulse of concentrated solution (320 g/L) each time. The feeding solution composition in pulse feed experiments and profile feed experiment was the same. In case of pulse feed, same feed solution was split into 2 parts of 250 ml each. Fed-batch experiments instead used an exponential profile ranging from 8 to 12 g/L·h⁻¹. Feed solution was strategically fed into bioreactors as per predefined profile. The latter started with flow rate of addition with glucose of 10 g/L·h⁻¹ for first hour. In next 2 h, the rate of glucose addition was increased to 12 g/L·h⁻¹. From 4th hour, feed rate was reduced to 10 g/L·h⁻¹ for another 2 h. From 6th hour, feed rate was maintained to be constant at 8 g/L·h⁻¹ till end of fermentation. The volume of feed solution was calculated based on expected volume of bioreaction broth and fed to controller system based on pre calibrated pump-pipe calculations.

Stock (about 20 OD at 600 nm) of *L. lactis* I7 was inoculated to 0.20 L of optimized medium in 0.25 L bottle and

Table 2 Component concentrations for salt OFAT experiments on small scale with constant glucose, soy peptone and yeast extract concentrations

Run No.	Glucose (g/L)	Soy peptone (g/L)	Yeast Extract (g/L)	Na ₂ HPO ₄ (g/L)	Di-Sodium Glycerophosphate (g/L)	K ₂ HPO ₄ (g/L)
1	20	20	15	19	NA	NA
2	20	20	15	NA	28.91	NA
3	20	20	15	NA	NA	30.54

incubated in a rotary air shaker (model Minitron Infors, Basel, Switzerland) at 30 °C and 150 rpm for 3–4 h. The inoculum size was 10% (v/v) and it was transferred to 2 L bioreactor to have initial optical density of 0.1 at 600 nm. All experiments were performed, at least, in triplicate.

High cell density experiments

The 2L bioreactor had been previously modified, by inserting four vertically set microfiltration modules to the baffles [18]. Each of the microfiltration modules used consisted of 10 capillaries of polypropylene Accurel PP, supplied by Sartorius (Melsungen, Germany) of 15 cm in length, with a diameter of 1.8 mm, a cut-off of 0.2 µm and a total filtering area of $1.76 \times 10^{-3} \text{ m}^2$. The microfiltration processes started 6–7 h after inoculation (2–3 h after the addition of concentrated feed). Filtrate exhaust medium was replaced with a fresh salt solution maintaining a constant volume of $2.0 \pm 0.1 \text{ L}$, using a level sensor. The experiments were performed in triplicate.

Downstream process using microfiltration and ultrafiltration membranes

As illustrated in the supplementary materials Figure S1, at the end of the fed-batch processes (23–24 h of growth), the supernatants were recovered after centrifugation at 6500 rpm for 30 min (Avanti j-20 XP, Beckman Coulter, Milan Italy), then microfiltered on 0.6 µm hollow fiber polyether sulfone membranes (Sartorius Stedim, Milan, Italy) to remove any cellular debris and/or aggregates. The supernatants were concentrated and diafiltrated by tangential flow filtration on a Sartoflow Alpha cross-flow filtration system, (Sartorius Stedim, Milan, Italy) on 10 kDa polyether sulfone membranes with 0.1 m^2 of surface area (Sartorius Stedim, Milan, Italy) and the permeates recovered were concentrated and diafiltrated on 2 kDa polyether sulfone membranes with 0.1 m^2 of surface area (Sartorius Stedim, Milan, Italy). The concentrated samples obtained (R10 kDa and R2 kDa) were used for antimicrobial activity.

Protein content

The total protein content was obtained using Bradford's method by analyzing the samples by UV/Vis spectrophotometry at 595 nm with a colorimetric method by using a protein assay kit from Bio-Rad Laboratories (USA) (Bradford, 1976) and the bovine serum albumin (BSA) as standard (Bio-Rad Laboratories, USA). A Beckmann DU-800 spectrophotometer was used.

Antimicrobial activity

Enterococcus faecalis ATCC 29,212™, *Salmonella enterica* subsp. *enterica* serovar *Typhimurium* ATCC® 14028GFP™ and enteroinvasive *Escherichia coli* (EIEC) ATCC 43,893™ were thawed and grown in tryptic soy broth (TSB) under

aerobic conditions for 24 h. In a 96-well plate, 1 µL of bacterial culture, 0.3 OD_{600} (approximately 10^8 CFU/ml), was added to 200 µL of TSB and 200 µL of M17 medium used as control sample. The same operation was accomplished, adding 1 µL of bacterial culture to 200 µL of concentrated sample 10 kDa and 2 kDa. The bacterial cultures were incubated at 37 °C aerobically for 24 h. At the end of the incubation, the bacteria were serially diluted, plated on tryptic soy agar (TSA) and incubated at 37 °C overnight for counting of colony forming units (CFU/mL). All experiments were performed in triplicate.

Lyophilization of biomass for stability studies

The fermented broth was centrifuged at 6500 rpm for 30 min to get biomass pellet. The wet weight of this biomass was measured. A trehalose sugar (solution of 50 g/L concentration) was added to this wet biomass. This composition was lyophilized for 24 h in lyophilizer (Christ Beta 2–8 LSCplus, Germany). After lyophilization, the powder was stored in sterile air-tight vessel at 4 °C for 12 months. A sample was taken out frequently in sterile conditions to check its viability. The sample was serially diluted using 0.9% NaCl solution up to presumed viability dilution factor. 10 µL was placed on M17 agar plates in triplicates. These plates were incubated at 30 °C for 24 h to get viability. All experiments were performed in triplicates.

Results

Small scale experiments: multiple factor screening using a Plackett-Burman screening (PBS) design

To identify potentially inter-dependent compounds within the medium recipe, a Plackett–Burman screening design was used. The detailed calculations are illustrated in the supplementary materials Table S1 and S2. Figures 1(a, b, c) report the effect of the components on biomass yield. Furthermore Fig. 1a indicates normal plot of the standard effects, and it gives a clear idea about ascorbic acid and MgSO_4 . One is negatively impacting the biomass growth and other one is not significantly helping biomass production respectively. Soya peptone products are the most effective nitrogen source, followed by yeast extract and tryptone. This is validated as shown in a Pareto chart of the standardized effects in Fig. 1b and main effects plot for biomass in Fig. 1c. Since soy peptone is performing better than tryptone, it was decided to only proceed with soy peptone. MgSO_4 is not significantly affecting the biomass production; thus, it was removed. On the other hand, ascorbic acid proved to reduce biomass production; therefore, it was removed.

The model also produced equation that expresses biomass production through mathematical model described as Eq. 3. $\text{Biomass} = 0.1886 + 0.01975 \text{ A} + 0.01609 \text{ B} + 0.01758 \text{ C} - 0.0669 \text{ D} + 0.0835 \text{ E} + 0.0165$.

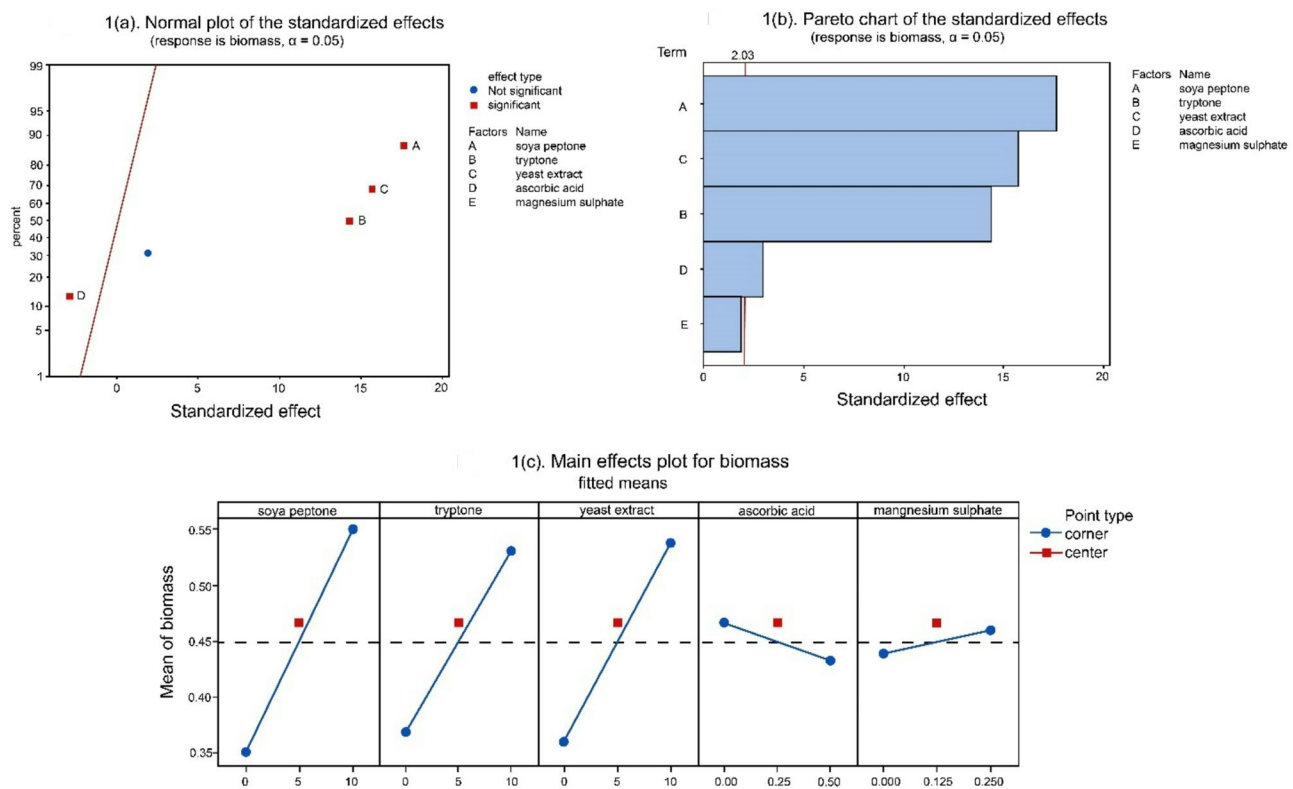


Fig. 1 (a) normal plot of the Standardized Effects, (b) factor wise Pareto chart of the standardized effect, (c) main Effects Plot for Biomass

Table 3 Effect of different phosphate sources: Na_2HPO_4 , $\text{C}_3\text{H}_7\text{Na}_2\text{O}_6\text{P}$, K_2HPO_4 with optical density at 600 nm on 0th, 6th and 24th hours of fermentation, lactic acid analysis in HPLC at 6th and 24th hour, LA productivity at 6th and 24th hours. Experiments were performed in 100 ml bottles with 80 ml volume to 30 °C with stirring of 150 rpm for 20 h

Salts	OD ₆₀₀ t0h	OD ₆₀₀ t6h	OD ₆₀₀ t24h	LA (g/L) t6h	LA (g/L) t24h	LA productivity (g/L·h) t0-6 h	LA productivity (g/L·h) t6-24 h
Na_2HPO_4	0.8±0.02	6.3±0.1	6.2±0.1	12.5±0.1	14.8±0.1	2.08	0.13
$\text{C}_3\text{H}_7\text{Na}_2\text{O}_6\text{P}$	0.5±0.03	5.6±0.1	5.6±0.2	10.5±0.1	14.1±0.1	1.75	0.21
K_2HPO_4	0.6±0.02	5.7±0.1	5.4±0.1	10.7±0.1	13.9±0.2	1.78	0.18

Table 4 Effect of different carbon sources (20 g/L) on the growth of *L. lactis* I7 with optical density at 600 nm on 0th, 6th and 24th hours of fermentation, lactic acid analysis in HPLC at 6th and 24th hours, LA productivity at 6th and 24th hours. Experiments were performed in 100 ml bottles with 80 ml volume to 30 °C with stirring of 150 rpm for 20 h

Carbohydrates	OD ₆₀₀ t0h	OD ₆₀₀ t6h	OD ₆₀₀ t24h	LA (g/L) t6h	LA (g/L) t24h	LA productivity (g/L·h) t0-6 h	LA productivity (g/L·h) t6-24 h
Glucose	0.3±0.1	6.9±0.2	5.8±0.1	19.2±0.3	21.3±0.5	3.2	0.11
Lactose	0.3±0.1	6.6±0.1	5.7±0.1	16.9±0.2	18.5±0.2	2.8	0.09

As shown in Fig. 1(b), yeast extract has effect of 17.61 on biomass production compared to 15.67 of soya peptone. Tryptone has effect of 14.34 against 2.03 baseline effect value at $\alpha = 0.05$. Ascorbic acid showed an effect of -2.98 and MgSO_4 have an effect of 1.86.

Small scale experiments: optimization of salts and sugars

In Tables 3 and 4, the results of different phosphate and carbon sources effects on growth of *L. lactis* I7 were reported.

The results show a reduction in optical density at 24 h compared to 6 h. The results also indicate almost 30 times higher LA productivity in the first 6 h compared to later 18 h (Table 4). Furthermore, as shown in Table 5, difference in fermentative growth between M17 reference

Table 5 Comparison between M17 media and optimized media for fermentation of *L. lactis* I7 with optical density at 600 nm on 0th, 6th and 24th hours of fermentation, lactic acid analysis in HPLC at 6th and 24th hours, LA productivity at 6th and 24th hours experiments were performed in 100 ml bottles with 80 ml volume to 30 °C with stirring of 150 rpm for 24 h

Carbohydrates	OD ₆₀₀ t0h	OD ₆₀₀ t6h	OD ₆₀₀ t24h	LA (g/L) t6h	LA (g/L) t24h	Residual carbon source (glucose/lactose) g/L	Initial pH	Final pH
M17	0.3±0.1	1.4±0.1	1.6±0.1	4.7±0.3	6.2±0.5	NA	6.9±0.2	5.2±0.1
M17 + Lactose (20 g/L)	0.3±0.1	2.7±0.2	2.4±0.3	9.8±0.1	11.1±0.3	12.3±0.2	6.9±0.1	3.9±0.1
M17 + Glucose (20 g/L)	0.3±0.1	3.2±0.2	2.6±0.3	10.5±0.2	12.3±0.3	11.4±0.1	6.8±0.1	3.9±0.2
						NA (lactose)		
Opt. media	0.3±0.1	4.6±0.1	5.4±0.1	12.5±0.2	14.2±0.2	3.7±0.3	7.1±0.2	3.9±0.1

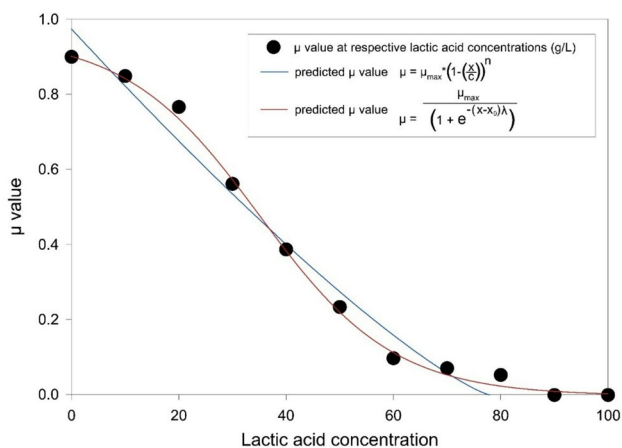


Fig. 2 Growth inhibition curve - μ was calculated for each lactic acid concentrations and graph was plotted against μ

media and optimized media can be clearly seen in terms of optical density and LA production at 6 and 24 h. Optical density at 6 h in optimized media was 228% higher than the growth in reference media. LA production during same period was observed to be 165% higher. Supplemented media also produced higher biomass and LA compared to reference media. Biomass optical density at 24 h for supplemented media runs were significantly lower than optical density at 6 h. This phenomenon is not observed for optimized media. It is clear that M17 media was able to reach pH where all metabolic activities for

L. lactis I7 would get inhibited. Based on final pH of fermented broth, it can be interpreted that supplementation of carbon sources successfully increased biomass and LA production.

Lactic acid tolerance

Use of Levenspiel model [27, 28] provided R^2 value of 0.9885, μ value of 0.9747 and n value of 1.2341 with c value of 78. But when we use Eq. 2, R^2 value reached 0.999, μ value of 0.9635 and X_0 value of 35.1748 with λ value of 0.08 (Fig. 2).

Bioreactor experiments

Batch processes were performed to verify the growth in fermenter using the optimized medium and parameters, comparing the results with the M17 medium.

In Table 6 all results obtained on bioreactor in the different processes (growth, viability, yield, and productivity) are reported. Optimized media almost provided double biomass, viability and LA production in batch processes. Pulse and profile fed-batch strategies produced equal amount of LA at the end of process, but there was slight reduction in biomass production for pulse fed-batch. On other hand, microfiltration experiments produced higher biomass density and 3 times higher viability. The calculations of LA produced per liter gets complicated due to total volume being different at different stage. *L. lactis* I7 growth, viability, cell density

Table 6 Summary of the results obtained in the different types of processes performed on 2 L bioreactor. *these values refer to the sum of lactic acid in the filtrate and inside the reactor. Experiments were performed in 2 L bioreactor at 30 °C and 150 rpm. Data were analyzed by two-tailed Non homoscedastic student's t-test. * indicates $p < 0.01$ compared to results obtained from batch experiments with M17 media; § indicates $p < 0.05$ compared to results obtained from batch with pulse experiments; §§ indicates $p < 0.01$ compared to results obtained from batch with pulse experiments

Type of process	medium	biomass (g/L)	viability (CFU/mL)	LA (g/L)	yield of the process (lactic acid produced/ glucose consumed)	productivity of LA (g/L-h)
Batch	M17	2.5±0.5	3.2±0.3·10 ⁹	7.6±0.5	0.45±0.10	0.35±0.02
Batch	optimized medium	5.4±0.3*	7.6±0.2·10 ⁹ *	15.8±0.5*	0.79±0.20*	0.72±0.05*
Batch with pulse	optimized medium	13.3±1.2*	1.1±0.1·10 ¹⁰ *	78.5±1.6*	0.95±0.20*	3.70±0.10*
Fed-batch	optimized medium	14.6±1.0*	1.9±0.1·10 ¹⁰ *§§	78.3±2.3*	0.96±0.15*	3.65±0.15*
High cell density/ Microfiltration	optimized medium	16.5±1.5*§	6.9±0.3·10 ¹⁰ *§§	77.8±0.5**	0.98±0.15*	3.66±0.02**

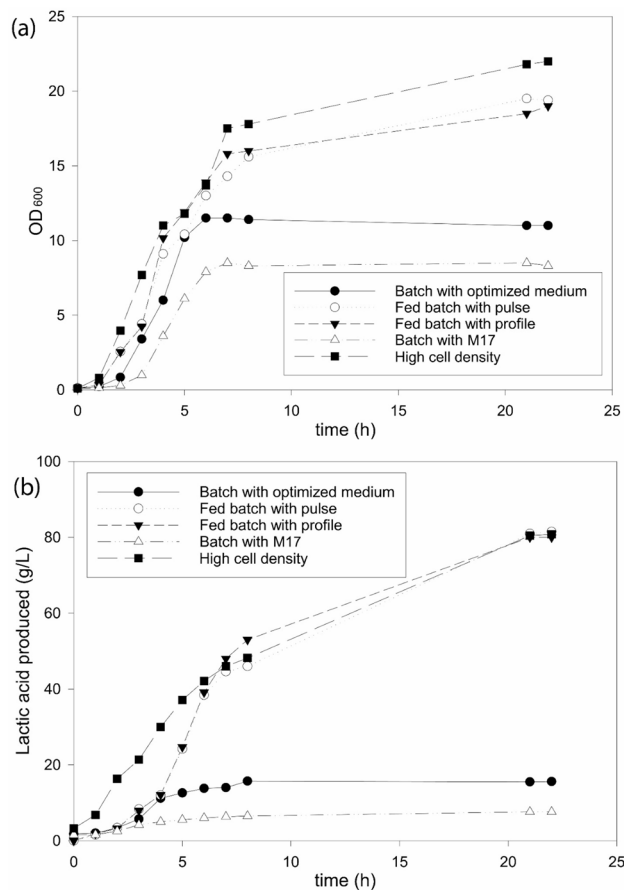


Fig. 3 Different types of bioreactor processes illustrating lactic acid production (a) growth curves of different types of process, (b) production of lactic acid in the different types of process

and LA production were also evaluated on different type of processes, as shown in Figs. 3(a) and 3(b).

The results obtained in the different fermentation processes show a clear significance when compared to the batch experiments with M17 medium used as a control. Furthermore, the viability data indicate an increase for the fed-batch processes, thanks to the optimization of an exponential profile and to the high cell density processes due to the continuous removal of growth inhibitory agents (such as lactic acid) compared to the batch with pulse processes.

Table 8 Results of antimicrobial activity: fold decrease in viability of three pathogenic strains grown *E. faecalis*, *S. Typhimurium*, and Enteroinvasive *E. coli* in M17 and TSB medium and with 2–10 kDa retentate solution

<i>E. faecalis</i>		<i>S. Typhimurium</i>		<i>EIEC</i>	
Sample	Fold decrease	Sample	Fold decrease	Sample	Fold decrease
TSB medium	10^1	TSB medium	10^1	TSB medium	10^1
M17 medium	10^1	M17 medium	10^1	M17 medium	10^1
Retentate 10 kDa Fermentation (protein content 0.52 mg/ml)	10^3	Retentate 10 kDa	10^4	Retentate 10 kDa	10^4
Retentate 2 kDa Fermentation (protein content 0.43 mg/ml)	10^6	Retentate 2 kDa	10^9	Retentate 2 kDa	10^8
LA content 5.2 mg/ml					
LA content 4.5 mg/ml					

Downstream process

Table 7 shows the most important parameters of ultrafiltration processes. The conductivity at the end of fermentation processes was around 50 mS/cm, thus the diafiltration was effective in reducing the electrolytes content (e.g. salts).

Flowrate was almost constant with a slight increase in transmembrane pressure (ΔP), the retentate of 2 kDa that should retain small proteins/oligopeptides with potential antibacterial effect was used for the analyses of bioactivity against pathogens, diafiltration was extensive to reduce as much as possible the influence of LA.

Antimicrobial activity

In Table 8 and in Fig. 4 we reported the results of 10 kDa retentate (R10 kDa) and 2 kDa retentate from 10 kDa

Table 7 Results of ultrafiltration processes: 10 kDa retentate and 2 kDa retentate with operational parameters as initial and final flux along with concentration factor, diafiltration volume and processed product analysis in terms of conductivity and LA

Process	initial transmembrane pressure (bar)	final transmembrane pressure (bar)	initial flux (LMH)	final flux (LMH)	conc. factor	dia-filtration volume	conductivity (mS/cm)	LA (g/L)
UF R10 kDa	0.30 ± 0.05	0.35 ± 0.05	12.5 ± 0.5	12.0 ± 0.5	13	3	16.50	15.2
UF R2 kDa	0.50 ± 0.05	0.80 ± 0.10	9.5 ± 0.5	8.0 ± 0.5	15	6	2.43	4.4

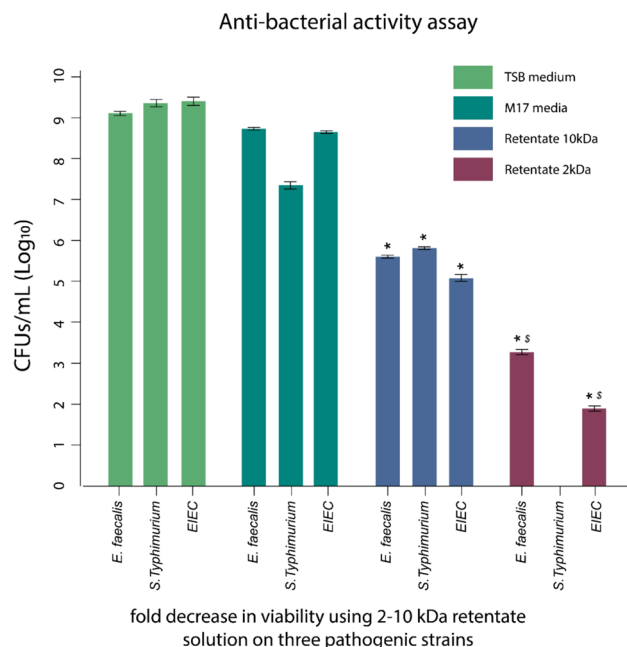


Fig. 4 Anti-bacterial assay activity in TSB medium, retentate 10 kDa, M17 media and retentate 2 kDa; fold decrease in viability using 2 and 10 kDa retentate solutions on three different pathogenic strains (*E. faecalis*, *S. Typhimurium*, and *Enteroinvasive E. coli*) Data were analyzed by two-tailed non homoscedastic Student's t-test. * indicates $p < 0.01$ compared to results obtained from control media (TSB and M17); § indicates $p < 0.01$ compared to results obtained using 10 kDa retentate

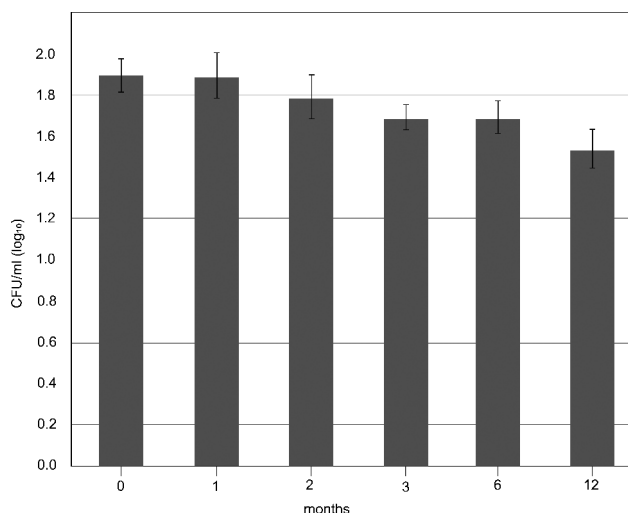


Fig. 5 Storage Stability (at 4 °C) of freeze-dried biomass. Viability was evaluated by resuspending the lyophilized powder at 10 mg/mL and inoculating on agar plated serial dilutions for 24 h for colony forming units counting

permeate (R2 kDa), which derived from the fed-batch processes. The results, obtained by an overnight incubation followed by a CFUs/ml counting, show the presence of a significant antibacterial activity on all tested pathogenic strains. In particular, the 2 kDa fraction showed a

reduction in pathogen viability between 10^6 and 10^9 logarithms compared to the control media.

Lyophilization of biomass for stability studies

The results of storage stability study (0–12 months) on viability of *L. lactis* I7 strain along with trehalose as cryopreservative are reported in Fig. 5. Averagely the decrease in viability was 15%, referred to initial CFU after the freeze-drying process.

Discussion

Traditional fermentation processes for cheese manufacturing in southern Italy are very often based on autochthonous microorganisms that are not well known or described. A starter consists of a mixture of living microorganisms, which are used to begin fermentation, producing specific changes in the properties of the food product [29]. In the framework of the BioNutra project (MUR ARS01_01166) we obtained the *L. lactis* I7 strain isolated from natural cow milk processing for the production of artisanal cheeses ripened without the addition of industrial starters cultures.

Probiotic potential and safety characteristics of this strain are under investigation with encouraging results in ongoing research. Besides the typical activity as starter (acidifiers/thickening agent etc.), we have evaluated a full array of metabolic features for this newly isolated strain.

The biomass production of starters and the metabolites of interest can be improved by optimizing growth parameters and specifically medium components [30]. For this reason, a large number of carbon and nitrogen sources beside salts, trace elements or micronutrients were screened. This is usually time and resource consuming due to the very high number of experimental runs needed. Tackling this issue from another perspective, one can consider a reference medium and evaluate its components for their effectiveness using a Plackett-Burman screening design (PBS design) to reduce the experimental runs. Once ineffective components are removed from the medium recipe, one factor at time (OFAT) can be used to replace existing medium components with significantly effective alternative sources. Diverse salts, and metal ions reported as effective in literature can be evaluated as additives (or to substitute others already used in the recipe). In fact, inclusion of monothetic methods for carbon and nitrogen screening simplifies the design process [31]. The approach proposed, namely PBS design followed by OFAT method using references, simplifies medium optimization procedures significantly [32]. Although this method will not yield the best possible results, it will provide significantly higher growth than the reference medium in a very short time and using fewer resources. Specifically, medium component screening can be achieved through PBS, Taguchi, and OFAT methods,

whilst medium component concentration optimization can be obtained through various responsive surface methodologies such as central composite design, Box-Behnken design, or full factorial design. Combination of traditional screening followed by a Plackett-Burman screening method for the design of experiments has been very common approach in the recent years [33, 34]. This paper focuses on the importance of optimization of DoE experiments to obtain a medium and identify a process to increase the growth of newly isolated *L. lactis* I7 and for metabolites production (i.e. lactic acid, bacteriocins). The model obtained from the DoE experiment is more than 95% accurate with alpha value of 0.05. The very high value of R^2 and R^2 -adjusted for the PBS model for screening of medium components indicates high accuracy of model and reliability in various conditions, respectively. The higher accuracy of the model provides confidence in model to screen-out components from the reference medium. In the following study di-sodium hydrogen phosphate has shown the ability to produce biomass significantly higher (12.5% higher than sodium glycerophosphate and 10.5% higher than K_2HPO_4) than other salt. The reason behind disodium hydrogen phosphate performing better than other salts is unknown. Since it was out of scope of the objective thus, it was not investigated further. In the experiments focusing on carbon sources, biomass production by glucose was not found to be significantly higher, but LA production was observed to be 13.6% higher during the first 6 h, showing a higher rate of carbon consumption. Since lactose is an animal-based source and there is a growing customer demand of lactose free products combined with vegan ones, glucose was favored over lactose use in bioprocess development. Simplified vegan grade medium contains glucose (20 g/L), soy peptone (20 g/L), yeast extract (15 g/L) and Na_2HPO_4 (19 g/L). The concentrations of each component were selected based on rational literature [32]. The media optimized on small scale was used for the experiments on 2L bioreactor in batch, batch with pulse, fed-batch and microfiltration mode. The results shown in Table 6 present an increase of biomass more than two-fold if we compare to a batch with M17 medium thanks to the DoE method. Ga-Hyun Choi and collaborators reported an increase in biomass of 1.58-fold for the standard medium (M17) and 1.85-fold under optimized cultivation conditions [33]. Supplemented media also produced higher biomass and LA compared to reference media. Biomass optical density at 24 h for supplemented media runs were significantly lower than optical density at 6 h. This phenomenon is not observed for optimized media. It is clearly evident that M17 media was able to reach pH where all metabolic activities for *L. lactis* I7 would get inhibited. Based on final pH of fermented broth, it can be interpreted that supplementation of

carbon sources successfully increased biomass and LA production. As indicated in Table 5, the newly developed media was able to produce almost 80% higher biomass compared to reference media and about 50% higher biomass than supplemented media (lactose or glucose). Optimized media produced 57% higher LA compared to reference media, 22% and 15% higher compared to supplemented media. It was also able to support higher growth (15.5%) compared to the integrated M17 media, further highlighting the advantages of the optimized formulation for higher biomass growth. Residual carbon present in fermented media indicates inhibition of growth possibly because of pH for supplemented and optimized media. But, in case of M17 reference media, growth stopped due to absence of sufficient carbon sources. Measuring pH values in final concluding experiments of small-scale experiments makes sure that factor of pH and residual substrate concentrations during optimization process were well negotiated. It is also evident that glucose is being consumed at a higher rate compared to that of lactose consumption, reaffirming glucose to be the first choice of carbon source for further experiments. This set of experiments provides vital information about the importance of optimized growth media against supplemented reference media. The media optimized on small scale was used for the experiments on 2L bioreactor in batch, batch with pulse, fed-batch and high cell density mode. Based on literature, 6.5 pH was considered to be the optimum pH for *L. lactis* I7 bioprocess development [35]. Once optimized media was developed, it was important to understand strain's ability to tolerate maximum possible LA concentration for biomass growth at given pH. The kinetic modeling of lactic inhibition study revealed more than 99% accuracy. This kinetic model (Eq. 2) was derived from experiments that were performed at the same initial pH, but final pH was not controlled. The final pH was dependent factor with respect to variable lactic acid concentration. Thus, even being a lactic acid inhibition model, it considers interaction between pH and lactic acid concentration in the growth phase and provides sigmoidal nature of curve (Fig. 2). Ann Zahle Andersen's pH model earlier predicts that a major cause of the decrease in the glycolytic rate, upon lowering the extracellular pH, is the lower pool of phosphoenolpyruvate available to fuel glucose uptake via the phosphoenolpyruvate-dependent transport system [36]. The inhibition caused by higher external lactic acid concentration is due to the solubility of the undissociated lactic acid within the cytoplasmic membrane and insolubility of dissociated lactate, which causes acidification of cytoplasm and failure of proton motive forces (PMF) [23]. Thus, it indicates combined effect of pH and lactic acid concentration on the PMF value resulting in inhibition of cellular growth, i.e., at different pH, bacterial

lactic acid tolerance will be different. It was demonstrated in alternative ways previously by Ten Brink et al. and Ryad A. et al. using different Eqs. [37, 38]. Ryad A. and co-authors. model was sigmoidal in nature and it was based on H^+ /Lactate against external pH values [38]. This modeling study helped in understanding the optimal value of glucose to be used in fed batch bioreactor experiments to maximize the LA yield. Based on this study, it was predicted that, biomass production will be totally inhibited around LA produced concentration of 79 ± 2 g/L. Thus, during bioprocess development through fed-batch, final total glucose in media was targeted to be 80 g/L. Based on this consideration, feed solution volume, and concentration of substrate was calculated. The lactic inhibition kinetic study also reveals us not to cross 12 g/L of LA concentrations during micro-filtration batch for ideal maximum possible biomass productivity. By comparing different growth methods used, it is clear that the addition of concentrated feed increases the yields and productivity in the fermentation processes, which are very similar even when changing the method of adding fresh substrate. In fact, Table 6 showed an increase of biomass production and viability, when we performed a fed-batch or batch with pulse. The yield of the process (LA produced/glucose consumed) is very high with the almost complete conversion of glucose into LA. The maximum productivity achieved was $15 \text{ g/L}\cdot\text{h}^{-1}$. The latter reduced significantly, when LA concentration in medium became superior to 50 g/L. This resulted in increased production and the productivity of LA through the feed processes by 5-fold respect to the batch processes. Hemalatha and collaborators demonstrated a 3.6-fold increase in production using statistical optimization by response surface methodology [39]. We subsequently investigated the use of the high cell density process. During fermentation processes the accumulation of a high quantity of LA in the broth inhibits the growth of LAB [18–40, 41]. Various processes have been proposed in the literature for the removal of LA, such as an electrodialysis system or co-cultures with lactate catabolizing yeast [42]. However, the former process is complex and expensive, and the ionic nutrients are also removed with LA. For this reason, our attention has been focused on the use of in situ product removal bioreactors which prevent the accumulation of LA and therefore growth inhibition by increasing biomass production. The efficiency of bioprocess is influenced by product inhibition due to the production of microbial metabolites in concentrations that become toxic even for the producing microorganism. Thus the in situ product removal technique appears as a strategy to overcome such problems by continuously removing the unwanted molecule as soon as it is produced [43]. As expected, this fermentation method increases the concentration of viable cells at the end of

the process, thanks to the removal of inhibitory agents during the process. In fact, viable cells increased by 0.5 and 1 log compared to the fed-batch and batch processes, respectively (Table 6). However, the more concrete reasons for using the high cell density process needs to be verified, given the higher costs and difficulties of the process, having obtained an increase of five to six times compared to batch processes with pulse and fed batch. The optimization of the medium, in addition to increasing the yields and productivity, will also reduce the costs of the process deriving from the raw materials used. Malvido and collaborators have demonstrated the possibility to obtain probiotic biomass with *L. lactis*, about 6 g/l with a viability of 2.1×10^{10} CFU/mL, and bacteriocins using whey as a culture medium supplemented with the main nitrogen sources present in the MRS medium (bacteriological peptone, meat extract and yeast extract) [11, 35]. In this work we propose the use of a fed-batch process for the production of bacteriocins with a medium free of animal sources obtaining approximately 15 g/L of probiotic biomass with a viability of 1.9×10^{10} CFU/mL. Furthermore, the use of glucose as a carbon source instead of lactose will allow production of vegan consumers friendly products. The supernatants obtained after fed-batch processes were ultrafiltered on 10 kDa and the permeate on 2 kDa membranes as shown in supplementary figure. A slight decrease in flow and increase in transmembrane pressure, of approximately 10–15%, can be noted, using 2 kDa membrane. This is due to concentration polarization phenomenon on the membrane surface and formation of the gel layer, during the concentration phase [45]. Subsequently a diafiltration phase was carried out to further eliminate salts, LA and low molecular weight proteins, further purifying the sample. The retentates of 10 and 2 kDa were used for antibacterial activity experiments. Although the results obtained (Table 8; Fig. 4) show a presence of significant antibacterial activity on all tested strains, differences were not noted between (among) supernatants harvested from different fermentation processes (i.e. batch with pulse, fed batch or high cell density). These results align with data already reported in literature, in which it has been demonstrated that other *L. lactis* strains were able to inhibit the growth, invasiveness, and production of virulence factors of these three species [45–48]. These characteristics are very important in the control of food spoilage, and human gut infectious diseases caused by *S. Typhimurium*, in bacillary dysentery caused by EIEC [50], and infections by *E. faecalis*. The latter reports of vancomycin-resistant enterococcus (VRE) strains in the late 1980s, was considered a harmless if not helpful, commensal species of the human gastrointestinal (GI) tract, on the contrary in recent times it was appointed among the major causes of hospital-acquired infections [49]. Finally, the stability (at 4 °C) of

the microbial lyophilizates over one year storage, that account for a decrease of about (15–20% colony forming units per milliliter further support the possibility of using the developed process to obtain *L. lactis* I7 dry powder to be used as a starter in traditional cheese manufacturing (Fig. 5), also considering the beneficial effects of its metabolites.

Conclusions

In this research work, an integrated biotechnological process has been designed for the new isolated *L. lactis* I7 strain describing an optimized vegan grade medium for fermentation and the enrichment of a partially purified fraction with antimicrobial activities. In the framework of this study, a modeling approach to identify medium components to increase biomass, LA and bioactive molecules production has been developed. In fact, carbon and nitrogen source regulation effect cell growth LA and bacteriocin production. A detailed component screening using Plackett–Burman and OFAT models showed that most of the salts and trace elements generally present in traditional media, like M17, are not affecting biomass production, thus a simplified medium preparation, with glucose, yeast extract, soy peptone, di-sodium hydrogen phosphate has been proposed. The results showed an increase of biomass of more than two-fold for batch with optimized medium with respect to traditional one, and six-fold increase for fed-batch processes. The in situ product removal process further increases the concentration of viable cells and the yields. However, process design and economics should be better studied to support technology transfer and scale-up at industrial level. The cultivation of *L. lactis* I7 on the optimized medium in batch and fed-batch process showed antimicrobial activity in the fractionated and partially purified supernatant in a range of MW as selected by ultrafiltration between 2 and 10 kDa. This fraction proved active against all the food pathogens tested.

Abbreviations

LA	Lactic acid
LAB	Lactic acid bacteria
DoE	Design of Experiments
HCDC	High cell density culture
UF	Ultra-filtration
DF	Diafiltration
rpm	Rotation per minutes
HPLC	High performance liquid chromatography
Kda	kilo dalton
OFAT	One factor at a time
v/v	Volume/volume
EIEC	Enteroinvasive Escherichia coli
OD ₆₀₀	Optical density
UF R10 kDa	Ultra filtration 10 kilo dalton retantate
UF R2 kDa	Ultra filtration 2 kilo dalton retantate
PMF	Protein motive force
MRS	De Man-Ragosa-sharp
CFU	Colony forming units

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12896-025-00997-z>.

Supplementary Material 1

Author contributions

AA conceptualization the project, writing-original draft and editing. AA and DP performed bioreactor, lyophilization experiments. DP performed Design of Experiments, data analysis, and numerical modelling, viability studies. IDC and LM performed isolation, classification, and nomenclature of strain. AF and VS performed antimicrobial activity test. LM, GD, and CS visualized project, project administration, writing original draft, review and editing. LM and CS performed funding acquisition, and supervision for the experimental activities.

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Data availability

Data is provided within the manuscript or supplementary information files.

Declarations

Ethical approval

This article does not contain any studies with human participants or animals performed by any of the authors.

Informed consent

Not applicable.

Institutional review board statement

Not applicable.

Competing interests

The authors declare no competing interests.

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